



《数值分析》14

主要内容:

样条插值的算例

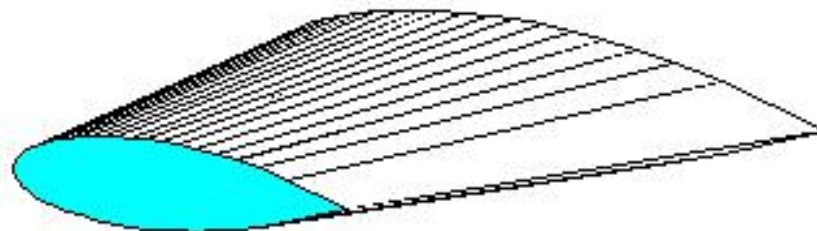
三次样条的概念

用一阶导数表示的样条

三次样条的极性

例1. 飞机机翼剖面图

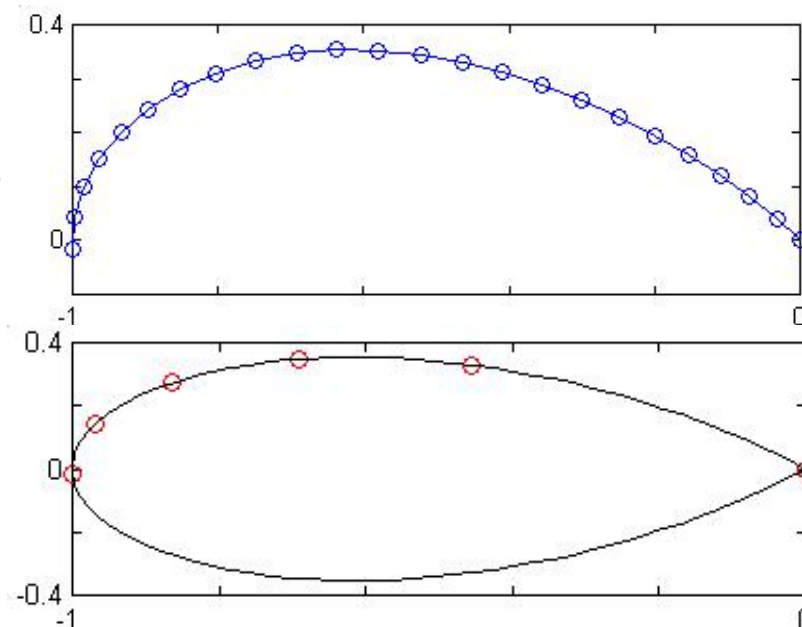
1. 数据采集



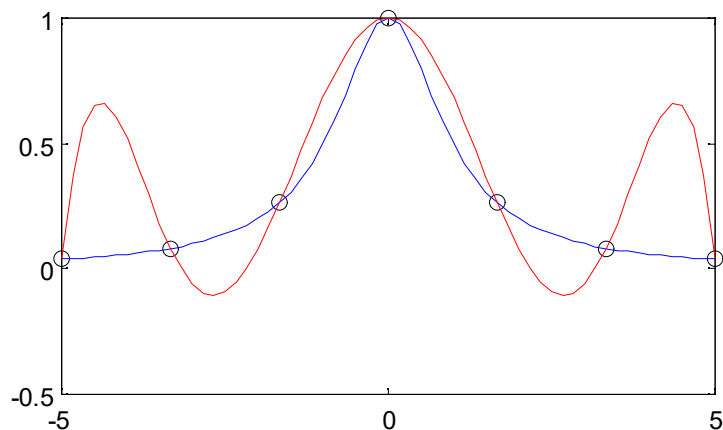
X	0	-0.4552	-0.6913	-0.8640	-0.9689	-0.9996
Y	0	0.3285	0.3467	0.2716	0.1408	-0.0160

2. 数据样条插值

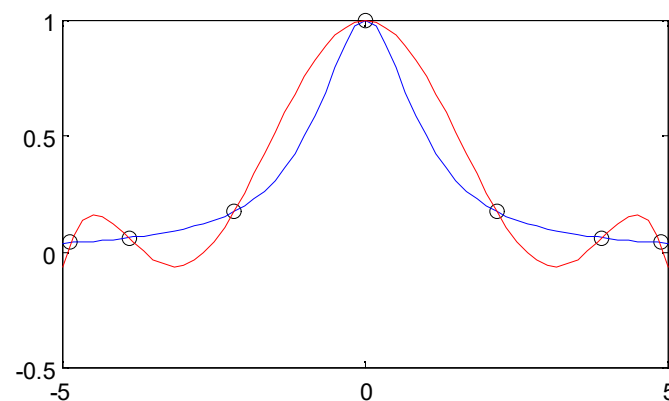
```
S=sqrt(diff(x).^2+diff(y).^2);  
S=[0,S];Sk=cumsum(S);  
sk=linspace(0,Sk(end),24);  
xt=spline(Sk,x,sk);  
yt=spline(Sk,y,sk);
```



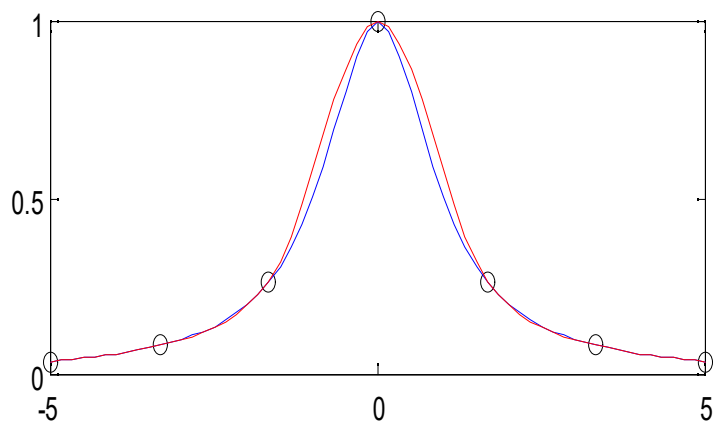
例2: 龙格函数的插值逼近 $f(x) = \frac{1}{1+x^2}$



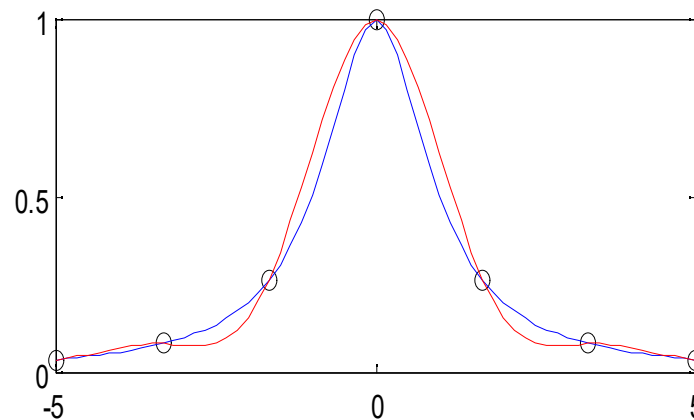
7结点等距插值



7结点切比雪夫插值



7结点埃尔米特插值



7结点样条插值

定义 5.4: 给定区间 $[a, b]$ 上的一个分划:

$$a = x_0 < x_1 < \dots < x_n = b$$

已知 $f(x_j) = y_j$ ($j = 0, 1, \dots, n$), 如果

$$S(x) = \begin{cases} S_1(x), x \in [x_0, x_1] \\ S_2(x), x \in [x_1, x_2] \\ \dots\dots\dots \\ S_n(x), x \in [x_{n-1}, x_n] \end{cases}$$

- 满足: (1) $S(x)$ 在 $[x_j, x_{j+1}]$ 上为三次多项式;
(2) $S''(x)$ 在区间 $[a, b]$ 上连续;
(3) $S(x_j) = y_j$ ($j = 0, 1, \dots, n$).

则称 $S(x)$ 为三次样条插值函数.

n 个三次多项式(每个三次多项式是4个待定系数), 待定系数共 $4n$ 个!!

当 $x \in [x_j, x_{j+1}]$ ($j = 0, 1, \dots, n-1$)时

$$S_j(x) = a_j + b_j x + c_j x^2 + d_j x^3$$

由样条定义,可建立方程 $(4n-2)$ 个!! **Why?**

插值条件: $S(x_j) = y_j$ ($j = 0, 1, \dots, n$)

连续性条件: $S(x_{j+0}) = S(x_{j-0})$ ($j = 1, \dots, n-1$)

$$S'(x_{j+0}) = S'(x_{j-0}) \quad (j = 1, \dots, n-1)$$

$$S''(x_{j+0}) = S''(x_{j-0}) \quad (j = 1, \dots, n-1)$$

- (1) 自然边界条件: $S''(x_0)=0, S''(x_n)=0$
- (2) 周期边界条件: $S'(x_0)=S'(x_n), S''(x_0)=S''(x_n)$
- (3) 固定边界条件: $S'(x_0)=f'(x_0), S'(x_n)=f'(x_n)$

例 5.7: 已知 $f(-1) = 1, f(0) = 0, f(1) = 1$. 验证下面分段三次多项式是自然样条函数.

$$S(x) = \begin{cases} \frac{1}{2}x^3 + \frac{3}{2}x^2, & x \in [-1, 0] \\ -\frac{1}{2}x^3 + \frac{3}{2}x^2, & x \in [0, 1] \end{cases}$$

证: 显然 $S(-1) = 1 \quad S(1) = 1 \quad S(0-) = S(0+) = 0$

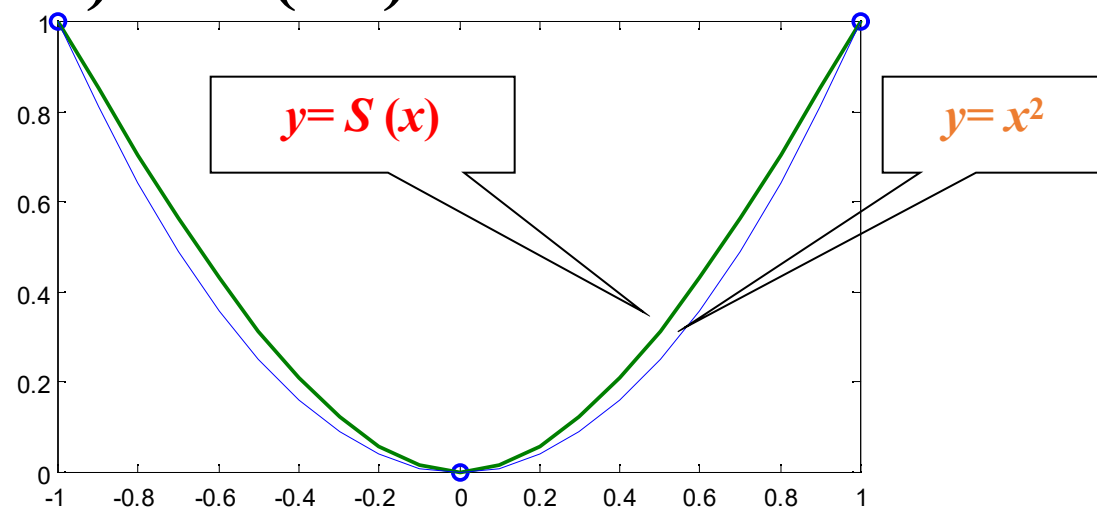
用一阶导数表示的样条

求导数得

$$S'(x) = \begin{cases} \frac{3}{2}x^2 + 3x, & x \in [-1, 0] \\ -\frac{3}{2}x^2 + 3x, & x \in [0, 1] \end{cases} \quad S''(x) = \begin{cases} 3x + 3, & x \in [-1, 0] \\ -3x + 3, & x \in [0, 1] \end{cases}$$

显然 $S'(-1) = -\frac{3}{2}$ $S'(1) = \frac{3}{2}$ $S'(0-) = S'(0+) = 0$
 $S''(-1) = 0$ $S''(1) = 0$ $S''(0-) = S''(0+) = 3$

所以, $S(x)$ 是满足插值条件且二阶导函数连续的分段三次多项式



分段Hermite插值公式导出的三次样条方法

已知函数表

x	x_0	x_1	$\cdots$	x_n
$f(x)$	y_0	y_1	$\cdots$	y_n

设 $f(x)$ 在各插值节点 x_j 处的一阶导数为 m_j (未知)

取 $x_{j+1} - x_j = h$, ($j = 0, 1, 2, \cdots, n$). 当 $x \in [x_j, x_{j+1}]$ 时,
分段Hermite插值

$$S(x) = \left(1 + 2\frac{x - x_j}{h}\right)\left(\frac{x_{j+1} - x}{h}\right)^2 y_j + \left(1 + 2\frac{x_{j+1} - x}{h}\right)\left(\frac{x - x_j}{h}\right)^2 y_{j+1} \\ + (x - x_j)\left(\frac{x_{j+1} - x}{h}\right)^2 m_j + (x - x_{j+1})\left(\frac{x - x_j}{h}\right)^2 m_{j+1}$$

由 $S''(x)$ 连续：有等式： $S''(x_j + 0) = S''(x_j - 0)$

考虑 $S''(x)$ 在区间 $[x_j, x_{j+1}]$ 和 $[x_{j-1}, x_j]$ 上表达式.

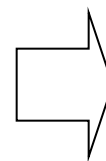
当 $x \in [x_j, x_{j+1}]$ 时, $S(x)$ 由基函数组合而成

$$\alpha_j(x) = \left(1 + 2\frac{x - x_j}{h}\right)\left(\frac{x_{j+1} - x}{h}\right)^2$$

$$\alpha_{j+1}(x) = \left(1 + 2\frac{x_{j+1} - x}{h}\right)\left(\frac{x - x_j}{h}\right)^2$$

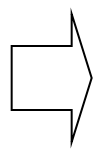
$$\beta_j(x) = (x - x_j)\left(\frac{x_{j+1} - x}{h}\right)^2$$

$$\beta_{j+1}(x) = (x - x_{j+1})\left(\frac{x - x_j}{h}\right)^2$$



$$\begin{cases} \alpha_j''(x_j) = \left[\frac{-8}{h^3}(x_{j+1} - x) + \left(1 + 2\frac{x - x_j}{h}\right) \frac{2}{h^2} \right]_{x=x_j} = -\frac{6}{h^2} \\ \alpha_{j+1}''(x_j) = \left[-\frac{8}{h^3}(x - x_j) + \left(1 + 2\frac{x_{j+1} - x}{h}\right) \frac{2}{h^2} \right]_{x=x_j} = \frac{6}{h^2} \end{cases}$$

$$\begin{cases} \beta_j''(x_j) = \left[\frac{4}{h^2}(x - x_{j+1}) + (x - x_j) \frac{2}{h^2} \right]_{x=x_j} = -\frac{4}{h} \\ \beta_{j+1}''(x_j) = \left[\frac{4}{h^2}(x - x_j) + (x - x_{j+1}) \frac{2}{h^2} \right]_{x=x_j} = -\frac{2}{h} \end{cases}$$



$$\begin{aligned} S''(x_j + 0) &= \alpha_j''(x_j)y_j + \alpha_{j+1}''(x_j)y_{j+1} \\ &\quad + \beta_j''(x_j)m_j + \beta_{j+1}''(x_j)m_{j+1} \end{aligned}$$

$$S''(x_j + 0) = -\frac{6}{h^2} y_j + \frac{6}{h^2} y_{j+1} - \frac{4}{h} m_j - \frac{2}{h} m_{j+1}$$

同理, 有

$$S''(x_j - 0) = \frac{6}{h^2} y_{j-1} - \frac{6}{h^2} y_j + \frac{2}{h} m_{j-1} + \frac{4}{h} m_j$$

联立得:

$$\begin{aligned} & -\frac{6}{h^2} y_j + \frac{6}{h^2} y_{j+1} - \frac{4}{h} m_j - \frac{2}{h} m_{j+1} \\ & = \frac{6}{h^2} y_{j-1} - \frac{6}{h^2} y_j + \frac{2}{h} m_{j-1} + \frac{4}{h} m_j \end{aligned}$$

→

$$m_{j-1} + 4m_j + m_{j+1} = \frac{3}{h}(y_{j+1} - y_{j-1})$$

($j=1, 2, \dots, n-1$)

设自然边界条件成立, 即

$$S''(x_0 + 0) = -\frac{6}{h^2} y_0 + \frac{6}{h^2} y_1 - \frac{4}{h} m_0 - \frac{2}{h} m_1 = 0$$

$$S''(x_n - 0) = \frac{6}{h^2} y_{n-1} - \frac{6}{h^2} y_n + \frac{2}{h} m_{n-1} + \frac{4}{h} m_n = 0$$

自然样条的导数值满足:

$$2m_0 + m_1 = \frac{3}{h}[y_1 - y_0]$$

$$m_{n-1} + 2m_n = \frac{3}{h}[y_n - y_{n-1}]$$

$$m_{j-1} + 4m_j + m_{j+1} = \frac{3}{h}(y_{j+1} - y_{j-1})$$

$$(j=1, 2, \dots, n-1)$$

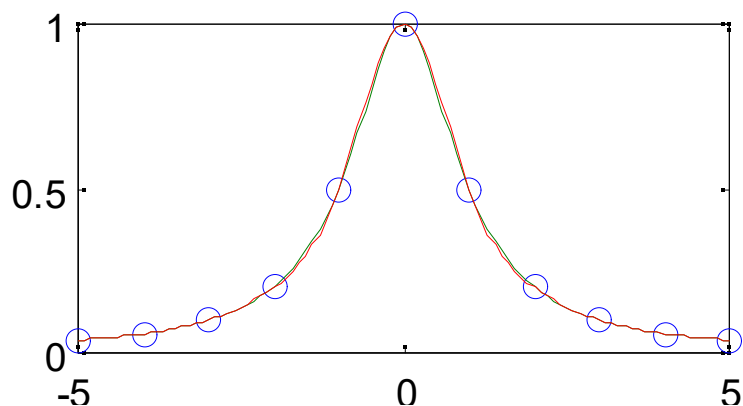
MATLAB样条命令: $y_i = \text{spline}(x, y, x_i)$

$x = -5:5; y = 1./(1+x.^2);$

$x_i = -5:0.1:5; f = 1./(1+x_i.^2);$

$y_i = \text{spline}(x, y, x_i); \text{error} = \max(\text{abs}(y_i - f))$

$\text{plot}(x, y, 'o', x_i, f, x_i, y_i, 'r')$



$\text{error} = 0.0220$

曲率比较: 计算公式

$$K = \frac{|y''|}{(1 + (y')^2)^{3/2}}$$

三次样条的极性

样条插值函数的极性

设 $f(x) \in C^2[a, b]$, 对于 $a = x_0 < x_1 < \dots < x_n = b$, 有 $f(x_j) = y_j (j=0, 1, \dots, n)$. $S(x)$ 是满足 $S(x_j) = y_j (j=0, 1, \dots, n)$ 的三次自然样条. 则有

$$\|S''(x)\| \leq \|f''(x)\|$$

$$\begin{aligned} \text{证明: } \|f''(x) - S''(x)\|^2 &= \int_a^b [f''(x) - S''(x)]^2 dx \\ &= \int_a^b [f''(x)]^2 dx - 2 \int_a^b f''(x) S''(x) dx + \int_a^b [S''(x)]^2 dx \\ &= \|f''\|^2 - \underline{2 \int_a^b [f''(x) - S''(x)] S''(x) dx} - \|S''\|^2 \end{aligned}$$

$$\begin{aligned}\int_a^b [f''(x) - S''(x)]S''(x)dx &= -\int_a^b [f'(x) - S'(x)]S'''(x)dx \\ &= -\sum_{j=1}^n S_j''' [f(x) - S(x)] \Big|_{x_{j-1}}^{x_j} = 0\end{aligned}$$

所以 $0 \leq \|f'' - S''\|^2 = \|f''\|^2 - \|S''\|^2 \quad \Rightarrow$

$$\|S''\|^2 \leq \|f''\|^2$$

即 $\int_a^b [S''(x)]^2 dx \leq \int_a^b [f''(x)]^2 dx$

样条函数 $S(x)$ 在 $[a, b]$ 上的总曲率最小.

样条插值的算例

三次样条的概念

用一阶导数表示的样条

三次样条的极性